

Does Potassium and Magnesium Supplementation Lower the Blood Pressure of Spontaneously Hypertensive Rats?

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Abstract □ The effects of an orally administered potassium and magnesium supplementation on the systolic blood pressure of 42 spontaneously hypertensive rats were determined for 10 weeks. The experimental preparation was mixed with standard rat pellets and this mixture was given to three rat groups at three concentrations: 6%, 9%, and 12%. Potassium- and magnesium-containing nutrient statistically significantly inhibited the normal increase in systolic blood pressure in these rats. It could even decrease the systolic blood pressure below the initial level and its effect was similar to the known calcium antagonist verapamil. The systolic blood pressure of rats fed with the standard pellet diet containing 6% sodium chloride (NaCl) increased from 191 to 209 mmHg, whereas that of animals on 6% K-Mg supplementation decreased from 197 to 181 mmHg. 9% and 12% K-Mg supplementation produced an almost equally significant decrease in systolic blood pressure, according to Student's unpaired *t*-test. The experimental nutrient could possibly be used in human studies on the feasibility of normalizing blood pressure by reducing the harmful effects of dietary salt.

High blood pressure is one of the main risk factors involved in heart attacks, stroke, and congestive heart failure.^{1,2} Elevated blood pressure affects a remarkable proportion of the population in industrialized countries. Therefore, safe and easily applicable methods are needed to combat high blood pressure in the community in general and also in hypertensive and hypertension-prone individuals in particular.

Epidemiological surveys and clinical studies support the finding that disturbances in calcium metabolism play a major role in both high blood pressure and atherogenesis.³⁻⁸ It is well-established that the condition is associated with a disturbance in the sodium and potassium transport mechanism in the cell membrane, and this is responsible for the increased resistance of peripheral blood circulation characteristic of hypertension.⁹⁻¹¹ Magnesium regulates the activity of the sodium pump in the cell membrane and hence plays an important role in the transport of sodium and potassium through the cell membrane.¹²⁻¹⁵ The acute effects of extracellular Mg²⁺ alterations appear to stem from sarcolemmal membrane interactions, including those of Ca-Mg, K-Mg, and Na-Mg.¹⁶ This has an effect on the state of contraction of the blood vessels and blood pressure via the sodium-calcium exchange mechanism.^{4,6,17}

In order to prevent cardiovascular disorders, increasing extracellular magnesium and potassium seems beneficial.^{2,18} Magnesium supplementation alone may cause a detrimental decrease in extracellular potassium, since the activated sodium-potassium-ATPase transports potassium into the intracellular space.¹⁹ Therefore, a potassium-magnesium combination is essential.

There is much evidence that decreasing sodium but increasing potassium and magnesium intake is beneficial in preventing and also treating high blood pressure.² But when used medically, especially potassium supplements can carry some risks. Adverse side effects such as hyperkalemia and a higher incidence of gastrointestinal mucosal lesions have been reported in susceptible

Table 1—Contents of the Experimental K-Mg Nutrient

Component	weight, mg	
Potassium chloride	589.8	respond K ⁺ 309.3 mg
Magnesium sulfate	173.1	respond Mg ²⁺ 50.9 mg
Microcrystalline cellulose	209.7	
Lysine hydrochloride	10.8	
Citric acid	5.4	
Glutamic acid	2.2	
Ascorbic acid	2.2	
Magnesium stearate	6.8	
Total	1000.0 mg	

humans as a result of salt substitutes.²⁰⁻²² Those especially vulnerable are elderly patients, patients taking potassium-sparing diuretics, patients with renal impairments, and patients taking nonsteroidal anti-inflammatory drugs.²³⁻²⁵ When supplements are administered to humans, care should be taken to monitor the potassium and magnesium plasma levels to minimize the risks. The form of the dosage is also important because it may be possible to develop an appropriate salt substitute to minimize the risks of adverse side effects.

To obtain further knowledge on the importance of balancing electrolyte status, we investigated the effect of an orally administered potassium and magnesium supplementation on the systolic blood pressure of spontaneously hypertensive rats. This model can suitably mimic human blood pressure.

Experimental Section

A total of 42 spontaneously hypertensive male rats, reared by Møllegaard ApS Breeding Center Ltd., Ejby, Denmark, were used in this study. At the beginning of the study, the 11-week-old rats were randomly divided into six subgroups of seven rats, and each subgroup was given a different diet. The sodium chloride diet was produced by adding 6% sodium chloride (E. Merck, Darmstadt, Germany) to the standard rat pellets (mice-rat fodder, Orion Corporation Ltd., Tuusula, Finland). The diet with potassium and magnesium supplements was produced by adding commercially available K-Mg preparation (Antisalt, SLK-companies, Helsinki, Finland) in three concentrations, 6%, 9%, and 12%, to the standard rat pellets. The verapamil diet was produced by adding 0.2% verapamil hydrochloride (verapamil hydrochloride, Orion Corporation Ltd., Espoo, Finland) to the standard rat pellets. The chemical composition of the potassium and magnesium preparation is presented in Table 1.

All the groups of rats except group 1 were fed with standard rat pellets mixed with different experimental ingredients. In all the groups of rats the average daily consumption of pellets was 20 g, and each group was allowed water ad libitum. The experimental design and abbreviations for each group of male rats are presented in Table 2.

The systolic blood pressure of the unanesthetized male rats was measured once a week for 10 weeks using a tail-cuff method and a blood pressure recorder (W + W electronic BP RECORDER 8002, Scientific Instruments Inc., 4000 Basel, Switzerland). Before the 10-week experimental measurement period the blood pressure of the 11-week-old rats was measured for 2 weeks in order to accustom the rats to the measurement techniques. Five consecutive measurements of the systolic

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